

RALPH WENZ FIELD, WY, USA

WMO: 720345

Lat: 42.796N

Lon: 109.807W

Elev: 2160

StdP: 77.93

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94086

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-23.9	-30.2	0.3	-26.4	-27.1	0.4	-23.4	12.9	-6.2	11.7	-5.7	1.8	300	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	29.9	12.4	27.7	11.6	26.7	11.3	14.3	24.5	13.2	22.8	12.4	22.6	4.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.8	11.3	15.9	10.2	10.1	14.5	8.8	9.2	14.5	47.7	25.1	44.2	22.4	41.8	22.1	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	9.4	8.3	DB	-32.8	31.3	2.9	1.1	-34.9	32.1	-36.6	32.8	-38.2	33.4	-40.4	34.2
			WB	-32.5	15.5	3.3	1.4	-34.9	16.5	-36.8	17.3	-38.6	18.1	-41.0	19.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-9.9	-8.7	-2.9	2.5	7.2	12.5	17.1	15.0	10.7	4.0	-2.7	-9.4
	DBStd	10.25	5.51	5.39	5.06	3.79	3.61	3.58	2.80	2.35	3.27	4.08	5.12	5.79
	HDD10.0	3084	617	524	398	226	101	17	1	1	31	187	381	602
	HDD18.3	5611	875	757	657	475	347	176	57	104	229	443	631	861
	CDD10.0	537	0	0	0	1	13	93	220	157	52	2	0	0
	CDD18.3	21	0	0	0	0	0	2	18	1	0	0	0	0
	CDH23.3	1388	0	0	0	0	5	198	740	370	73	2	0	0
	CDH26.7	317	0	0	0	0	0	42	219	51	5	0	0	0
Wind	WSAvg	3.1	2.8	2.8	3.1	3.7	3.4	3.4	3.1	3.0	2.9	3.0	3.0	2.6
Precipitation	PrecAvg	293	16	14	16	23	43	32	27	27	30	20	19	16
	PrecMax	492	48	39	38	60	103	100	76	79	89	58	52	49
	PrecMin	132	0	2	4	0	8	4	0	1	0	1	0	0
	PrecStd	76	11	9	10	13	27	25	19	16	19	14	12	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	6.0	14.8	19.9	23.9	30.3	33.9	29.9	27.4	21.9	13.9	6.1
		MCWB	0.0	1.1	5.1	7.1	8.7	12.1	14.3	11.5	11.1	8.7	4.7	1.3
	2%	DB	1.4	2.6	10.9	16.3	22.3	27.5	31.1	27.8	25.3	17.6	11.3	2.5
		MCWB	-0.8	0.1	3.5	5.3	8.5	11.1	13.2	11.0	9.7	7.1	3.8	-0.1
	5%	DB	0.1	1.2	7.3	13.8	20.1	25.9	28.9	27.1	23.7	15.8	7.9	1.1
		MCWB	-1.6	-0.8	2.2	4.4	7.9	10.6	12.6	11.0	9.3	6.2	2.4	-0.8
	10%	DB	-1.1	0.0	4.9	11.4	17.3	23.0	27.4	25.9	22.1	12.8	5.2	-0.2
		MCWB	-2.5	-1.6	0.9	3.8	7.1	9.8	11.9	10.8	8.7	5.1	0.9	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.8	1.9	5.6	7.7	10.2	13.2	17.9	14.1	13.2	9.7	6.1	1.9
		MCDB	1.9	4.4	12.9	18.3	19.9	24.8	22.9	20.8	21.3	16.3	13.2	3.9
	2%	WB	-0.2	0.7	4.0	6.2	9.3	12.3	15.6	13.0	11.9	8.4	4.3	0.5
		MCDB	0.9	2.4	9.7	14.3	19.2	23.8	26.5	21.2	19.5	14.9	9.4	1.9
	5%	WB	-1.4	-0.4	2.6	5.2	8.5	11.5	14.1	12.3	10.7	7.2	3.1	-0.7
		MCDB	-0.3	0.9	6.7	12.3	17.9	22.6	25.0	21.9	19.0	13.5	7.2	0.8
	10%	WB	-2.6	-1.6	1.2	4.1	7.7	10.7	13.1	11.7	9.8	5.9	1.5	-2.0
		MCDB	-1.3	-0.4	4.2	10.7	16.1	21.1	23.1	22.3	18.8	12.0	5.0	-0.7
Mean Daily Temperature Range	5% DB	MDBR	13.9	13.6	12.7	13.3	15.1	17.9	19.7	19.9	18.7	14.8	13.5	13.3
		MCDBR	13.2	12.2	14.4	18.9	21.1	22.2	22.2	22.7	23.2	19.8	16.5	12.0
		MCWBR	11.1	9.8	8.3	9.0	9.0	7.8	7.0	8.4	9.3	9.9	9.9	9.4
	5% WB	MCDBR	12.3	11.3	13.2	16.2	18.7	19.4	18.1	18.4	18.3	16.7	14.5	11.2
		MCWBR	10.7	9.3	7.9	8.4	8.4	7.3	6.1	7.1	8.0	8.7	8.8	9.2
Clear-Sky Solar Irradiance	taub	0.225	0.235	0.248	0.291	0.304	0.304	0.326	0.320	0.290	0.260	0.236	0.223	
	taud	2.399	2.380	2.445	2.404	2.466	2.506	2.466	2.475	2.554	2.608	2.555	2.453	
	Ebn at Noon	945	988	1010	979	968	964	939	937	950	951	933	922	
	Edh at Noon	75	91	98	112	108	105	108	103	88	73	64	65	
All-Sky Solar Radiation	RadAvg	2.11	3.03	4.35	5.43	6.08	7.29	7.21	6.24	5.03	3.38	2.23	1.70	
	RadStd	0.14	0.26	0.28	0.34	0.52	0.55	0.37	0.26	0.40	0.25	0.17	0.13	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page